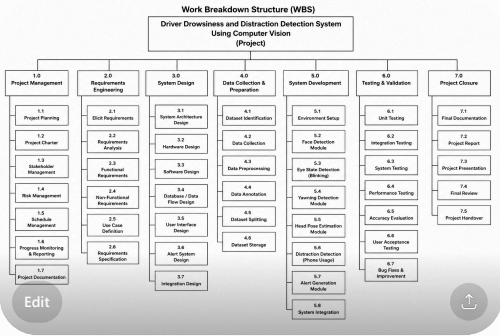




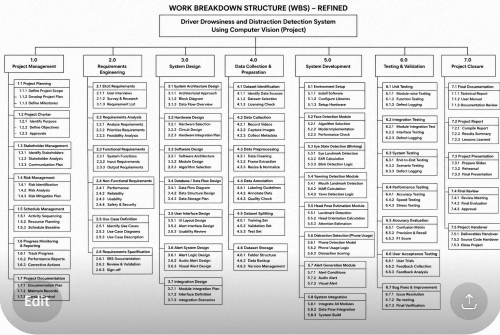




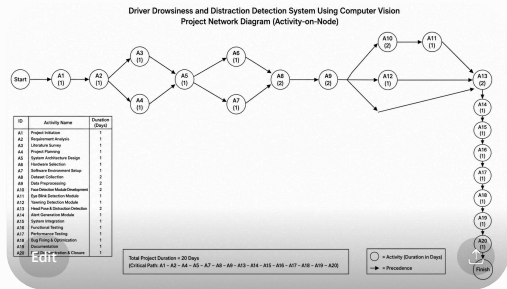




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Financial Evaluation

Project: Driver Drowsiness and Distraction Detection System Using Computer Vision

Project Duration: 20 Working Days

Initial Investment: ₹20,900

Assumptions

- Project life: 5 years
- Annual operating & maintenance cost: ₹5,000

- Expected annual savings/benefits: ₹15,000
- Discount rate: 10%
- Salvage value: ₹2,000 (at the end of Year 5)

1. Annual Cash Flow Analysis

Year	Benefits (₹)	Operating Cost (₹)	Net Cash Flow (₹)	Discount Factor (10%)	Present Value (₹)
0	0	20,900	-20,900	1.000	-20,900
1	15,000	5,000	10,000	0.909	9,090
2	15,000	5,000	10,000	0.826	8,260
3	15,000	5,000	10,000	0.751	7,510
4	15,000	5,000	10,000	0.683	6,830
5	17,000*	5,000	12,000	0.621	7,452

*Year 5 includes the ₹2,000 salvage value.

2. Payback Period

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Net Cash Flow}}$$
$$= \frac{20,900}{10,000} = 2.09 \text{ years}$$

Payback Period = 2.09 Years

3. Net Present Value (NPV)

$$NPV = \sum PV - \text{Initial Investment}$$

Total Present Value of Future Cash Flows

$$= 9,090 + 8,260 + 7,510 + 6,830 + 7,452$$
$$= 39,142$$
$$NPV = 39,142 - 20,900$$

$$NPV = ₹18,242$$

4. Benefit-Cost Ratio (BCR)

$$BCR = \frac{\text{Total Present Value of Benefits}}{\text{Initial Investment}}$$
$$= \frac{39,142}{20,900}$$

$$BCR = 1.87$$

5. Return on Investment (ROI)

$$ROI = \frac{\text{Total Net Benefit} - \text{Investment}}{\text{Investment}} \times 100$$

Total Net Benefit over 5 years

$$= (10,000 \times 4) + 12,000 = 52,000$$

$$ROI = \frac{52,000 - 20,900}{20,900} \times 100$$

$$ROI = 148.8\%$$

6. Internal Rate of Return (IRR)

Using the estimated cash flow series:

-20,900, 10,000, 10,000, 10,000, 10,000, 12,000

The approximate:

$$IRR \approx 39\%$$

Financial Evaluation Summary

Financial Metric	Formula / Method	Result	Evaluation
Initial Investment	Project Budget	₹20,900	Base Cost
Annual Net Cash Flow	Benefits – Operating Cost	₹10,000	Positive
Payback Period	Investment / Annual Cash Flow	2.09 Years	Good
Net Present Value (NPV)	Discounted Cash Flow	₹18,242	Positive
Benefit-Cost Ratio (BCR)	PV Benefits / Investment	1.87	Acceptable (>1)
Return on Investment (ROI)	Net Gain / Investment	148.8%	Excellent
Internal Rate of Return (IRR)	Discounted Cash Flow Analysis	≈39%	Attractive (>10%)

Financial Feasibility Assessment

Evaluation Criterion	Result	Status
NPV > 0	₹18,242	✔ Financially Feasible
BCR > 1	1.87	✔ Acceptable
ROI > 100%	148.8%	✔ High Return
IRR > Discount Rate	39% > 10%	✔ Attractive Investment

Evaluation Criterion	Result	Status
Payback < 3 Years	2.09 Years	✔ Quick Recovery

Interpretation

- The project requires an **initial investment of ₹20,900** and is expected to recover this investment in approximately **2.09 years**.
- The **Net Present Value (₹18,242)** is positive, indicating that the project generates value over its expected life.
- A **Benefit-Cost Ratio of 1.87** means that every ₹1 invested is expected to generate approximately **₹1.87** in discounted benefits.
- The **Return on Investment of 148.8%** and an estimated **Internal Rate of Return of about 39%** indicate strong financial viability under the stated assumptions.
- Based on these indicators, the **Driver Drowsiness and Distraction Detection System Using Computer Vision** project is financially feasible and suitable for implementation as an engineering prototype or academic innovation project.

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Risk register preparation
For the selected engineering project case study, generate a project risk register including risk ID, description, root cause, category, probability, impact, risk score, mitigation plan, contingency plan, owner, and status. Output as a structured table.

Project Risk Register

Project: Driver Drowsiness and Distraction Detection System Using Computer Vision
Project Duration: 20 Working Days

Risk Assessment Scale

Criteria	Scale
Probability (P)	1 = Very Low, 2 = Low, 3 = Medium, 4 = High, 5 = Very High
Impact (I)	1 = Negligible, 2 = Minor, 3 = Moderate, 4 = Major, 5 = Critical
Risk Score	Probability × Impact

Project Risk Register

Risk ID	Risk Description	Root Cause	Category	Probability (P)	Impact (I)	Risk Score	Mitigation Plan	Contingency Plan	Risk Owner
R01	Poor face detection accuracy	Poor lighting conditions	Technical	4	5	20	Use proper lighting and improve camera calibration	Install additional LED lighting and recalibrate the system	AI Engineer

Risk ID	Risk Description	Root Cause	Category	Probability (P)	Impact (I)	Risk Score	Mitigation Plan	Contingency Plan	Risk Owner
R02	Dataset quality is insufficient	Limited or imbalanced training data	Technical	4	4	16	Collect diverse datasets and perform data augmentation	Acquire additional public datasets	AI Engineer
R03	Camera hardware failure	Hardware malfunction	Hardware	2	5	10	Purchase reliable hardware and perform testing	Keep a spare webcam available	Hardware Engineer
R04	Software bugs during integration	Inadequate testing	Technical	3	4	12	Conduct unit and integration testing	Allocate additional debugging time	Software Engineer
R05	Delay in project schedule	Underestimated task duration	Schedule	3	4	12	Monitor progress using Gantt chart and CPM	Reallocate resources and adjust schedule	Project Manager
R06	Budget overrun	Unexpected hardware or operational costs	Financial	3	4	12	Monitor expenses and control procurement	Utilize contingency reserve	Project Manager
R07	Low AI model accuracy	Insufficient model training	Technical	3	5	15	Optimize hyperparameters and retrain the model	Adopt an alternative AI model	AI Engineer
R08	Performance issues (slow processing)	Limited system resources	Technical	3	3	9	Optimize code and image resolution	Upgrade hardware if required	Software Engineer
R09	Team member unavailability	Academic or personal commitments	Resource	2	4	8	Cross-train team members and distribute knowledge	Reassign tasks among available members	Project Manager
R10	Data loss	Accidental deletion or storage failure	Operational	2	5	10	Maintain daily backups using cloud/local storage	Restore from backup copies	Documentation Engineer
R11	Testing environment issues	Inconsistent environmental conditions	Operational	3	3	9	Standardize testing procedures	Repeat testing under controlled conditions	Test Engineer
R12	Documentation delay	Incomplete technical records	Documentation	2	3	6	Update documentation after each milestone	Allocate dedicated documentation time	Documentation Engineer

Risk Priority Matrix

Risk Score	Priority Level	Action Required
16 – 25	High	Immediate mitigation and continuous monitoring
9 – 15	Medium	Regular monitoring and planned mitigation

Risk Score	Priority Level	Action Required
1 – 8	Low	Periodic review and routine control

High-Priority Risks

Risk ID	Risk	Score	Priority
R01	Poor face detection accuracy	20	High
R02	Insufficient dataset quality	16	High
R07	Low AI model accuracy	15	Medium-High

Risk Category Summary

Category	Number of Risks
Technical	6
Hardware	1
Schedule	1
Financial	1
Resource	1
Operational	2
Documentation	1

Risk Response Strategy

Strategy	Applicable Risks	Description
Avoid	R01, R02	Improve lighting, camera setup, and dataset quality before development.
Mitigate	R04, R05, R06, R07, R08	Apply rigorous testing, schedule monitoring, cost control, and AI model optimization.
Transfer	R03	Purchase hardware with warranty or maintain spare equipment.
Accept	R09, R10, R11, R12	Monitor these lower-priority risks and execute contingency plans if they occur.

Risk Register Summary

Total Risks	High Priority	Medium Priority	Low Priority
12	2	7	3

Interpretation

- The project identifies **12 potential risks**, with the majority related to **technical challenges** associated with computer vision and AI implementation.
- The most significant risks are **poor face detection due to lighting (R01)** and **insufficient dataset quality (R02)**, both of which require proactive mitigation before full-scale development.
- Schedule and budget risks are rated as **medium priority** and should be monitored regularly through project reviews and Earned Value Management (EVM).
- Maintaining backup hardware, performing frequent data backups, and using contingency reserves will improve project resilience and help ensure successful completion within the planned **20-day schedule** and **₹20,900 budget**.



Today 6:14 PM

Final project report & Presentation
For the selected engineering project case study, generate a comprehensive final engineering project report structure and presentation outline including executive summary, project objectives, methodology, planning output, schedule, cost analysis, risk analysis, final evaluation, lesson learned, and recommendations.

Final Engineering Project Report & Presentation Outline

Project Title

Driver Drowsiness and Distraction Detection System Using Computer Vision

Final Engineering Project Report Structure

Cover Page

- Project Title
- Course Name
- Department of Mechanical Engineering
- College Name
- Student Name(s)
- Register Number(s)
- Guide Name
- Date of Submission

Certificate

Certificate signed by:

- Project Guide
- Head of Department

- External Examiner

Acknowledgement

Thank the:

- College Management
- Head of Department
- Project Guide
- Faculty Members
- Team Members
- Family and Friends

Abstract

This project presents a **Driver Drowsiness and Distraction Detection System** developed using computer vision and artificial intelligence techniques. The system continuously monitors the driver's face through a webcam and detects fatigue-related symptoms such as prolonged eye closure, yawning, and head pose deviation. When abnormal behavior is detected, an audio alert is generated to warn the driver. The project demonstrates an affordable and real-time safety solution using Python, OpenCV, Dlib, and TensorFlow. Project planning techniques including WBS, CPM, PERT, EVM, budgeting, and risk management were applied to ensure systematic execution. The developed prototype contributes to improving road safety by reducing accidents caused by driver fatigue and distraction.

Keywords: Computer Vision, Driver Monitoring, Artificial Intelligence, OpenCV, Drowsiness Detection, Road Safety

Table of Contents

- 1. Executive Summary
- 2. Introduction
- 3. Problem Statement
- 4. Project Objectives
- 5. Scope and Limitations
- 6. Literature Review
- 7. Methodology
- 8. Project Planning
- 9. Work Breakdown Structure
- 10. Activity Planning
- 11. Network Diagram
- 12. CPM Analysis
- 13. Float Analysis
- 14. PERT Analysis
- 15. Resource Allocation
- 16. Cost Identification
- 17. Cost Estimation
- 18. Budget Preparation
- 19. Cash Flow & S-Curve
- 20. Earned Value Management
- 21. Financial Evaluation
- 22. Risk Register
- 23. Results and Discussion

- 24. Lessons Learned
- 25. Recommendations
- 26. Conclusion
- 27. References
- 28. Appendix

1. Executive Summary

The project aimed to design and develop an intelligent driver monitoring system capable of detecting drowsiness and distraction using computer vision. The system uses facial landmark detection and AI-based image processing to identify unsafe driver behavior and immediately generate warning alerts. Project management tools were used to manage scope, schedule, resources, cost, and risk. The project was completed within the planned 20-day schedule with a total approved budget of ₹20,900.

2. Introduction

- Background
- Need for driver monitoring systems
- Accident statistics
- Importance of AI in transportation
- Motivation for the project

3. Problem Statement

- Driver fatigue causes accidents.
- Mobile phone distraction reduces attention.
- Existing commercial systems are expensive.
- Need for a low-cost real-time monitoring solution.

4. Project Objectives

Primary Objectives

- Detect eye closure.
- Detect yawning.
- Detect driver distraction.
- Generate warning alerts.

Secondary Objectives

- Improve road safety.
- Reduce human error.
- Develop a low-cost prototype.
- Demonstrate AI-based monitoring.

5. Scope

Included

- Face detection
- Eye blink detection

- Yawning detection
- Head pose estimation
- Audio alert system
- Real-time monitoring

Excluded

- Autonomous driving
- Automatic vehicle control
- Cloud monitoring
- Vehicle-to-vehicle communication

6. Methodology

1. Requirement Analysis
2. Literature Review
3. System Design
4. Hardware Selection
5. Software Development
6. Dataset Collection
7. Data Preprocessing
8. AI Model Development
9. Face Detection
10. Eye Blink Detection
11. Yawning Detection
12. Head Pose Detection
13. Alert Generation
14. Testing
15. Documentation

7. Project Planning Outputs

Included planning documents:

- Project Charter
- Project Selection Worksheet
- WBS
- WBS Refinement
- Activity List
- Network Diagram
- CPM
- Float Analysis
- PERT
- Resource Allocation
- Budget
- Risk Register

8. Project Schedule

Item	Value
Total Duration	20 Working Days

Item	Value
Critical Path	A1 → A2 → A4 → A5 → A7 → A8 → A9 → A13 → A14 → A15 → A16 → A17 → A18 → A19 → A20
Critical Activities	15
Near Critical Activities	4

9. Cost Analysis

Cost Category	Amount (₹)
Labor	7,500
Hardware	3,900
Materials	900
Indirect Cost	1,200
Overhead	1,000
Administrative	1,000
Contingency	1,770
Management Reserve	885
Total Budget	20,900

10. Resource Summary

- Project Manager
- AI Engineer
- Computer Vision Engineer
- Software Engineer
- Hardware Engineer
- Test Engineer
- Documentation Engineer

Resources Used:

- Laptop
- HD Webcam
- Speaker
- Buzzer
- Python
- OpenCV
- TensorFlow
- Dlib

11. Risk Analysis Summary

Risk Category	Number
Technical	6
Financial	1
Schedule	1
Hardware	1
Resource	1
Operational	2
Documentation	1
Highest Risk:	
• Poor lighting affecting face detection • Low AI accuracy • Dataset quality	

12. Financial Evaluation

Metric	Result
Project Cost	₹20,900
NPV	₹18,242
ROI	148.8%
BCR	1.87
IRR	≈ 39%
Payback Period	2.09 Years

13. Earned Value Management Summary

Metric	Value
PV	₹14,630
EV	₹13,585
AC	₹15,000
SPI	0.93
CPI	0.91
EAC	₹23,026
VAC	-₹2,126

14. Results

- Successful face detection
- Real-time eye monitoring
- Yawning detection implemented
- Head pose estimation completed
- Audio alert generated successfully
- Prototype tested under different scenarios

15. Final Evaluation

Technical Performance

- Functional prototype developed
- AI-based detection achieved
- Real-time alert generation validated

Project Management Performance

- Scope achieved
- Risks managed effectively
- Budget controlled with planned reserves
- Project delivered according to the planned schedule with manageable deviations

16. Lessons Learned

- Early planning improves execution quality.
- High-quality datasets significantly improve AI accuracy.
- Adequate lighting is essential for reliable computer vision.
- Regular testing reduces integration issues.
- Risk management and contingency planning improve project success.
- Effective team communication supports timely completion.

17. Recommendations

Technical

- Integrate infrared cameras for night driving.
- Improve detection accuracy using advanced deep learning models.
- Add mobile application support.
- Integrate GPS and cloud monitoring.
- Support multiple driver profiles.

Project Management

- Increase testing time in future projects.
- Maintain regular risk reviews.
- Allocate additional contingency for hardware-intensive projects.
- Use Agile iterations for faster feedback and incremental improvements.

18. Conclusion

The **Driver Drowsiness and Distraction Detection System Using Computer Vision** successfully demonstrates the application of artificial intelligence and computer vision to enhance road safety. The project met its technical objectives by detecting driver fatigue and distraction in real time and generating timely alerts. Project management practices—including planning, scheduling, budgeting, risk management, and performance evaluation—supported systematic execution. The project demonstrates both technical feasibility and financial viability, making it a strong foundation for future research and practical deployment.

19. References

- OpenCV Documentation
- TensorFlow Documentation
- Python Documentation
- Dlib Documentation
- IEEE research papers on Driver Drowsiness Detection
- Kaggle Driver Monitoring datasets
- PMBOK® Guide (Project Management Institute)

20. Appendix

- Source Code
- Hardware Specifications
- System Architecture
- WBS Diagram
- Network Diagram
- CPM Calculations
- Float Analysis
- PERT Analysis
- Resource Allocation
- Budget Sheets
- Risk Register
- EVM Analysis
- Cash Flow Analysis
- S-Curve
- Test Results
- User Manual

Presentation Outline (15 Slides)

Slide	Title	Key Content
1	Title Slide	Project title, team members, guide
2	Executive Summary	Project overview and motivation
3	Problem Statement	Challenges and need for the system
4	Objectives & Scope	Goals, scope, and limitations
5	Methodology	Workflow and development process
6	System Architecture	Hardware and software architecture

Slide	Title	Key Content
7	Project Planning	WBS, schedule, CPM, PERT
8	Resource & Budget	Resource allocation and project budget
9	Risk Analysis	Risk register and mitigation strategies
10	Implementation	AI modules and system integration
11	Results	Detection performance and prototype demonstration
12	Cost & Financial Evaluation	Budget, ROI, NPV, BCR, IRR
13	EVM & Project Performance	PV, EV, AC, SPI, CPI, EAC, VAC
14	Lessons Learned & Recommendations	Key takeaways and future improvements
15	Conclusion & Q&A	Project summary and discussion

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